ADNOC GAS OPERATIONS & MARKETING LLC

GOLDEN WELD PROCEDURE

EPCm CONTRACT NO.: 3000076501
ADNOC GAS PROJECT NO.: 30201-50150-LT-HE

LONG TERM - EXECUTE STAGE

PROJECT 5.0 (P5)

EPCm FOR HABSHAN DBN

A	ISSUED FOR COMPANY REVIEW (ICR)	21-JUL-25	LITHIN Construction Engineer	PRADEESH Principal Construction Engineer	GARY Construction Manager
REV.	DESCRIPTION	DATE	PREPARED	CHECKED	APPROVED
			wood.		

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1.0 INTRODUCTION

1.1 BACKGROUND

ADNOC aims to increase oil production to produce 5.0 MMBOPD by year-end 2027. Accordingly, associated gas production is anticipated to increase. Also, to meet the future oil and gas demand, non-associated gas developments are considered. Hence, the overall gas supply scenarios for major ADNOC Gas (hereafter referred to as COMPANY) assets i.e., Das Facilities, Bab and Habshan Complex (Habshan 0, 1, 2, 4, and 5), Asab and Bu Hasa, is anticipated to change depending on their feed sources, profiles including rates, compositions, and operating conditions. These changes will also subsequently impact the COMPANY downstream facilities, particularly, products handling and transferring facilities i.e., Sales Gas Network, NGL Pipeline Network, Ruwais Fractionation Trains (1, 2, 3, 4 and upcoming new Train 5 part of Bab Cap Gas SELECT Study), Sulphur Handling Facilities, and Condensate systems, etc.

COMPANY has carried out the Long Term DEFINE for the project with M/s Amec International Ltd. (M/s Wood) and subsequently decided to proceed with Long Term EXECUTE phase execution to ensure the readiness of COMPANY facilities to accommodate increased gas supply by December 2027.

Accordingly, COMPANY has awarded EPCm contract to M/s Amec International Ltd. (M/s Wood) to carry out the Long Term scope requirements. M/s Wood shall act as a single entity in the performance of the services and a single point of accountability to ADNOC Gas for the successful completion of Long Term EXECUTE phase.

1.2 LONG TERM EPCM PROJECT OBJECTIVES

The main objective of the Long Term EPCm is to execute the Habshan DBN Project to support ADNOC to achieve oil production target of 5.0 million BOPD so that Associated and Non-Associated Gas can be processed within the overall processing capacity of Habshan. The Long Term EPCm scope includes the following major areas as a minimum:

1. Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in)
2. Bab CDS and Bab NGL
3. Habshan 5
4. Pipeline scope activities including below new pipelines
   I. Bab CDS to Habshan Complex HP pipeline
   II. Bab CDS to Habshan Complex MP pipelines (2 no’s)
   III. Habshan 5 to Habshan 2 pipeline

The scope also includes adequacy checks for Ruwais NGL Fractionation Trains, MERAM,

Habshan Sulphur Granulation Plant, Sales gas, NGL, Condensate and interconnecting Pipeline network and Utilities for all impacted AG facilities.

The Long Term EPCM scope includes the following major activities as a minimum:

1. Detailed Engineering and design services
2. Procurement Services
3. Construction Management services
4. Pre-commissioning, Commissioning, Start-up services
5. Final documentation handover and Project close-out

Project also involves major interface activities with other on-going projects. Refer Project Interface management plan (Doc Num: PP-30201_50150_LT_HE-GG-000-50-00-006) for complete list of interface projects.

1.3 SCOPE BATTERY LIMIT

Refer Design Basis (Doc Num: BOD-30201_50150_LT_HE-GG-000-00-00-001) for the scope battery limits, Gas allocation and detailed Design basis.

2.0 PURPOSE OF THIS DOCUMENT

This Procedure defines the process for tracking and management of closure welds at P5 HABSHAN EPCM Project.

3.0 ORDER OF DOCUMENT PRECEDENCE

In the event of any conflict between the applicable documents, the order of precedence shall be as follows:

1. UAE Statutory Legislation and Environmental Regulations
2. ADNOC HSE Regulations, HSE Standards and HSE Code of Practice
3. Data Sheets
4. Design Basis
5. Project Specifications and addendum to AGES / DGS (Note-1)
6. COMPANY Standards / Specifications and AGES / DGS (Note-1)
7. International Codes and Standards

Note-1 AGES Addendum/AGES shall be considered wherever AGES addendum/AGES is available. DGS Addendum/DGS shall be considered only in case AGES Addendum/AGES is not available.

However, in case of conflict between the various specifications, standards and the

requisition, most stringent requirements shall govern. EPCm CONTRACTOR/VENDOR shall refer any such conflict, and any other matter, which requires further interpretation, to the attention of COMPANY and COMPANY's interpretation shall be final. Resolution of any conflict shall be obtained from COMPANY in writing before proceeding.

Any deviation from COMPANY standard accepted in the existing facility design shall not be considered as basis for this PROJECT design and any deviations shall be submitted to COMPANY for approval before proceeding.

4.0 REFERENCE, CODES AND STANDARDS

4.1 REFERENCE DOCUMENTS

Table 4-1: Reference Documents

Sl. No.	Document Number	Document Title
1.	30201_50150_LT_HE-ADD-SP-07-007	ADDENDUM TO AGES-SP-07-007 WELDING AND NON-DESTRUCTIVE EXAMINATION (NDE)
2.	30201_50150_LT_HE-ADD-6300-004	ADDENDUM TO DGS-6300-004 PIPELINE WELDING SPECIFICATIONS
3.	PP-30201_50150_LT_HE-GG-000-51-00-001	PROJECT QUALITY PLAN
4.	PP-30201_50150_LT_HE-GG-000-51-00-006	QUALITY REQUIREMENTS FOR CONSTRUCTION CONTRACTOR
5.	PP-30201_50150_LT_HE-GG-000-51-00-004	CONSTRUCTION QUALITY PLAN
6.	30201_50150_LT_HE-ADD-SP-09-001	ADDENDUM TO AGES-SP-09-001 PIPING BASIS OF DESIGN
7.	30201_50150_LT_HE-ADD-SP-09-007	ADDENDUM TO AGES-SP-09-007 SPECIFICATION OF SHOP AND FIELD FABRICATION OF PIPING
8.	SP-30201_50150_LT_HE-GG-000-13-00-001	SPECIFICATION FOR BOLT TORQUING TENSIONING PROCEDURE FOR FLANGED CONNECTIONS

4.2 ADNOC STANDARDS

Table 4-2: ADNOC Standards

Sl. No.	Document Number	Document Title
1.	AGES-GL-13-001	Contractors QAQC requirement specification
2.	Contract - Annexure 14	Quality Control and Integrity Management
3.	AGES-SP-13-001	Criticality rating specification
4.	AGES-SP-13-002	Procurement Inspection and certification requirements in projects

4.3 INTERNATIONAL CODES AND STANDARDS

Table 4-3: International Codes and Standards

Sl. No.	Document Number	Document Title
1.	ASME B31.3	Process Piping
2.	ASME IX	Qualification Standard for Welding, Brazing and Fusing Procedures
3.	AWS D1.1	Structural Welding Code Steel
4.	ASME VIII	Rules for construction of pressure vessels

5.0 DEFINITIONS

5.1 GENERAL DEFINITIONS

For this document, the following general definitions are applicable.

Table 5-1: General Definitions

DEFINITION	DESCRIPTION
COMPANY	ADNOC GAS (AG) for whom the SERVICES are being performed by the EPCm CONTRACTOR.
EPCm CONTRACTOR	Amec International Limited – Abu Dhabi (WOOD) appointed by COMPANY to perform Engineering, Procurement, Construction management, Pre-Commissioning and Commissioning stage for Long Term Solution (under EXECUTE phase).

DEFINITION	DESCRIPTION
FACILITIES	Habshan Complex (H0, H1 Train 1/2, H2, H4, H3 for SS-300 Tie-in), Bab CDS, Bab NGL, Habshan 5 and interconnecting pipelines.
LICENSOR	Supplier of packaged units and responsible for the operating and design performance.
MANUFACTURER/ VENDOR/ SUPPLIER	The Party (parties) which manufactures and/or supplies material, equipment, technical documents or drawings and/or services to perform the duties specified by the EPCm CONATRCATOR
PMC	Project Management Consultant appointed by COMPANY for the PROJECT
PMT	Project Management Team
PROJECT	PROJECT 5.0 (P5), Long Term EPCm for HABSHAN DBN
SERVICES	The scope of work to be completed during the EXECUTE stage (EPCm) of the PROJECT.
SHALL	The use of the word “shall” indicate a mandatory requirement
SHOULD	The use of the word “should” indicate a strong recommendation to comply with the requirements of this document
EPCm CONSTRUCTION CONTRACTOR	Subcontractor appointed by EPCm CONTRACTOR to carry out Construction works at site for respective construction packages.
SUB-CONTRACTOR	The COMPANY approved party, which carries out services subcontracted by EPCm CONTRACTOR.

5.2 TECHNICAL DEFINITIONS

For this document, the following technical definitions are applicable.

Table 5-2:Technical Definitions

DEFINITION	DESCRIPTION
Acid Gas	A gas stream containing predominantly H_2S and/or CO_2 from an Acid Gas Removal Unit
Condensate	A hydrocarbon liquid stream containing Pentane+ compounds
Dehydrated Gas	A gas stream that has been dehydrated to remove water

DEFINITION	DESCRIPTION
Dry Gas	A gas stream containing no liquids at flowing conditions, but which may still contain water and hydrocarbons that will fall out if there is a change in flowing conditions.
Feed Gas - Associated Gas	A Gas stream that is produced from an oil well during the extraction of crude oil.
Feed Gas – Non-associated Gas	Hydrocarbon gas originating from gas reservoirs.
GMP 2023	Gas Master Plan 2023
Lean Gas	Lean nature of gas is defined based C2+ in the gas stream. If C2+ is less than 15% then stream is defined as Lean gas
Mixed Feed Gas Stream	Mixed associated and non-associated gas stream
NGL	A liquid stream with primarily C2, C3, C4 and some C5+ compounds.
Rich Gas	Rich gas is defined based C2+ in the gas stream. If C2+ is more than 15% it is considered rich gas
Sales Gas	Residue gas sent to the Sales gas network.
Sour Gas	A gas stream containing significant amounts of H_2S and/or CO_2.
Sustainable Profile (GMP 2023)	GMP 2023 High Supply Scenario – unconstrained oil production from GMP Guidelines Document from all associated gas fields
Sweet Gas	A gas stream in which the H_2S and CO_2 are not present in significant amounts.
Technical Profile	AON provided profile for technical case for associated gas.
Technical Optimized case	The ADNOC Gas facilities (Das, Asab-0/3, Buhasa, Ruwais, associated pipelines) will be sized to accommodate technical rates from associated gas fields.

DEFINITION	DESCRIPTION
	Habshan facilities will be sized to accommodate technical rate from all associated gas fields (Bab, Buhasa, NEB, Offshore) but not simultaneously. All inlet facilities / inlet compression shall be designed for technical rate.
Unconstrained Production Profile / High Supply Scenario	This shall be based on Up stream's OPEC Unconstrained Oil production case. This scenario corresponds to sustainable projects profiles
Closure Weld	The final weld connecting piping systems or components which have been successfully pressure tested in accordance with the provisions of ASME B31.3, ASME B31.8 . Welds designated as Closure Welds are not planned to be pressure tested, prior to mechanical completion
Golden Weld	A slang term sometimes used to indicate a Closure Weld. This term has been used, but is neither descriptive nor part of the terminology used by the governing Code of Construction.
MCWL	Master Closure Weld Log
NCR	Non-Conformity Report
GTAW	Gas Tungsten Arc Welding
SMAW	Shielded Metal Arc Welding

5.3 ABBREVIATIONS

ADNOC	Abu Dhabi National Oil Company
AGES	ADNOC Group Engineering Specifications
CSU	Commissioning and Start-Up
D&A	Decommissioning & Abandonment
DGS	Design General specification
DEFINE	Front End Engineering Design (FEED)

EPC	Engineering Procurement & Construction
EPCM	Engineering Procurement & Construction Management
FEED	Front End Engineering Design
HSE	Health, Safety and Environment
IFA	Issued For Approval
IFR	Issued For Review
QA/ QC	Quality Assurance/ Quality Control
RFI	Request for Information

6.0 RESPONSIBILITIES

6.1 EPCM CONSTRUCTION CONTRACTOR

6.1.1 Project Field Engineer

The Project Field Engineer is responsible for the following:

a) Establishing the requirements and execution strategy of the site-specific Closure Weld Management Procedure.

b) Implementing the requirements of this procedure.

c) Ensuring that site supervision, craft and field engineering personnel involved in the execution of closure welds have reviewed and understand this procedure and the associated requirements.

d) Ensuring that field engineering personnel have required training and competency

e) to execute their responsibilities in accordance with this procedure.

6.1.2 Welding Engineer

a) Assigning a Welding Engineer to carry out in-process inspection of a Closure Weld upon receipt of a signed permit required to this job.

b) Liaises with the Piping Superintendent to confirm the assignment of the welding engineer and to verify timing of required in-process inspection.

c) Ensure all welding engineers are trained, competent, and qualified to execute the inspection requirements stipulated by this procedure.

d) Performing in-process examination in accordance with ASME B31.3, paragraph 344.7.

e) Documenting in-process examination in accordance with ASME B31.3, paragraph 344.7

f) EPCm Contractor and Company shall be notified thro' "Request for inspection" prior to the commencement of a Golden Weld.

f) Shall verify all NDE procedure were approved by EPCm Contractor and Company.

g) PAUT can be used in lieu of Radiography through EPCm Contractor and COMPANY approval only. If PAUT is used, verify the procedure validation shall be done on a test piece with known diameter and thickness and defect size and locations and Witnessed by Third Parity Inspection.

6.2 EPCM CONTRACTOR

6.2.1 Project Field Engineer

The Project Field Engineer is responsible for the following:

a) Obtaining the golden joint approval from Company

b) Preparation of necessary documentation for the golden joint approval

6.2.2 Site QC Engineer / Welding Engineer

The Site QC Engineer / Welding Engineer is responsible for the following:

a) Verify the approval for golden joint in place from COMPANY.

b) Verify ISO and PID drawings related to the concerned golden welds, line numbers, and joint numbers.

c) Verify that the Welding Procedure Specification (WPS) and Procedure Qualification Record (PQR) are approved and available.

d) Verify the Welder Qualification Records and ensure the welder is qualified for the specific welding procedure and material.

e) Confirm that the base materials and filler materials meet the specified requirements.

f) Assign a Welding inspector to carry out in-process inspection of a closure weld upon receipt of a signed permit required to this job.

EPCM CONTRACTOR Inspector responsibilities include but not limited to;

- Review and approval for closure golden weld joint Location.
- Verify the approved WPS/PQR/WPQT.
- Visual inspection on the fit up and preparation before welding
- Visual inspection on each pass
- Carry out final weld visual inspection. If accepted release for NDT.
- Carry out inspection activities as per Sec 7.3 of this procedure.

7.0 PROCEDURE

7.1 JUSTIFICATION

a) Pressure testing will be planned and used to the maximum extent reasonable and the number of welds designated as a closure weld shall be minimized.

b) closure welds shall be witnessed by COMPANY.

7.2 GENERAL

7.2.1 Golden joints agreement

GOLDEN Joints (closure welds) shall be pre-identified, and a golden joints summary register shall be prepared with all information, including line number, class, diameter, thickness, location, joint number, type of welding, and markings on the related line drawings, and submitted to the COMPANY for approval, no later than once the work package construction starts.

7.2.2 CUTTING & BEVELING

Pipe end preparation shall be in accordance with approved welding procedure as specified in marked up isometric, using oxy-fuel, plasma, mechanical or manual cutting/grinding tools.

7.2.3 ERECTION & ASSEMBLY

Butt weld faces to be joined shall be aligned in accordance with approved welding procedure (WPS).

a) Welding summary control sheet shall be maintained by the QC Engineer.

b) All closure welds shall be marked /identified on the isometric drawing and included inside the piping test package.

c) Requests for in-process inspection shall be submitted at least 24 hours in advance of a closure weld being carried out.

d) Prior to performing any closure weld a permit form shall be completed and signed by the Piping engineer This shall be included in closure weld documentation package.

e) All NDE, PHWT, and Hardness Tests when required, shall be performed in accordance with ASME B31.3, Design code and Project Specifications. All documentation shall be completed and submitted to DCC.

f) Completed closure welds shall be recorded in welding summary control sheet

g) In the event that a closure weld is performed without a welding engineer present to carry out an in-process visual inspection, the weld shall be cut out and re-welded with the required in-process inspection which also includes tack welds.

7.3 INSPECTION

a) All inspections shall be carried out to ASME B31.3, Design code and Project Specification

requirements.

b) all Joints bevels shall be checked via MPI before fit up.

C) Joint to be MPI or PT after hot pass prior to continue filling and CAP.

b) All completed closure welds will be subject to 100% (radiography & 100 % manual UT), 100% MPI, in case of PAUT accepted in lieu of RT then golden joint shall be 100% PAUT & 100% MUT)

c) All NDE, including In-Process Visual Examinations, associated with a Closure Weld shall be documented and traceable to the specific weld using the In-Process Visual Examination Report, which shall be included in the closure weld documentation package.

d) In process welding and NDT examinations for each and every Closure Weld shall be witnessed by EPCM CONSTRUCTION CONTRACTOR, EPCM CONTRACTOR & COMPANY the welding inspector & NDT in accordance with ASME B31.3 paragraph, 344.7, In-process.

e) any interruption in welding shall be after 40 % completion of the joint.

f) in case of any weld interruption MPI shall be carried and preheat before restart welding

e) Examination, to include the following:

- Joint preparation and cleanliness.
- Preheating.
- Fit-up, joint clearance and internal alignment prior to joining i.e. any type of welding inclusive of tack welding.
- Visual inspection shall be performed for each pass
- Variables specified by the joining procedure, including filler material.
- Condition of the root pass after cleaning – external and, when accessible, internal – aided by liquid penetrant or magnetic particle examination
    - Slag removal and weld condition between passes.
- Appearance of the finished joint.

f) Each of the above in-process examination points shall be noted as either acceptable, rejected or N/A for each closure weld performed.

7.4 CLOSURE WELD DOCUMENTATION

All reports and records associated with closure weld activity, including any repairs and associated NDE, shall be included in the document package. Each package shall include, as a minimum:

- An isometric drawing showing the closure weld, its location, and its weld ID number.
- Justification for the use of the closure weld designation
- Welding procedure(s) used for the closure weld.
- Completed and signed leak (pressure) test records for the adjoining piping if applicable
- Cleanliness Records
- Individual NDE records for complete NDE.
- The originals closure welds inspection record.

APPENDIX-01 GOLDEN WELD REPORT

wood.	P5 HABSHAN EPCM PROJECT	sheet	Rev.	DATE
	PROJECT NO.	1 of 1	0	
	Closure (Golden) Weld Report	Report Number:		

Test Package Numbers									
Dwg/ISO Number							Line Class		
Weld Information									
Weld Number		Dia.		W. T		Joint Type		Weld Position	
Preheat		Interpass.				PWHT Req'd?			
WPS						Filler Material			
Welder ID						Material Type			

In-Process Examination as required by ASME B31.3

Attribute	Values	Accept/Reject	FEW Initial
Joint Preparation and Cleanliness			
MPT Check			
UT Check			
Fit-up: Internal misalignment criteria for butt welds			
Socket Weld Gap verified with scribe during fit up	maximum 1/16"		
Preheat if required by WPS	1.5mm – 3.0mm		
Filler Material (record classification)	Root Fill		
Visual Examination of Root Pass			
PT Examination of Root Pass			
Visual Examination of Fill Passes			
PT Examination of Fill Passes			
Visual Examination of Completed Weld			
PT Examination of Completed Weld			
NDE of Final Weld (100% VT + MPT + UT+RT+ Hardness for butt welds)			
Verification Documents			
PWHT Chart Number (if applicable)			
RT Report Number (Required for Butt Welds Only)			
MPT/PT Report Number			
VT Report Number			
UT Report Number			
Hardness Report Number			

	Contractor Welding Engineer	Company Welding Engineer
NAME		
DATE		
SIGNATURE		

APPENDIX-02 MASTER GOLDEN WELDS LOG

wood.	P5 HABSHAN EPCM PROJECT	ADNOC		
		sheet	Rev.	DATE
	PROJECT NO.	1 of 1	0	
Master Golden Welds Log (MGWL)				

Ser	Weld No.	Isometric No.	Approved by ADNOC (DATE)	Final Visual Examination (DATE)	NDT & Hardness Completion	TIE IN Completion
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						